Basel Shrief Hemaid

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Portfolio: Basel Shrief

EDUCATION

Faculty of Engineering, Cairo University – Cairo, Egypt — ***Sep 2023 – Expected July 2028***

B.Sc. in Electronics and Communication Engineering.

Specialized in designing IC's and Software for Embedded Systems according to the global standards.

Work Experiences & Summer Trainings

- Embedded Software Engineering Paid Intern | Sparrow — ***Mar 2026 – till now***
  Led the end-to-end embedded software development of the Sparrow system, building firmware modules, integrating ESP-based hardware peripherals, communication interfaces, unit testing workflows, Flutter mobile application connectivity, while delivering feature upgrades, system validation, debugging, and performance improvements in a paid internship environment.
- RTOS Embedded Cairo University Eco Racing Team member with full technical Documentation — ***Aug 2025 – till now***
- Networks and Embedded Training over the TBM with PetroJet in Project new Cairo Metro line 4 — ***July 2025 – Aug 2025***
- Facilitated dialogue between students and faculty as Class Representative, directly impacting curriculum adjustments; efforts resulted in a 20% increase in course satisfaction ratings as reported in end-of-semester surveys. — ***Aug 2024 – Aug 2025***

ACHIEVEMENTS

- **Visited Poland Shell Eco-marathon —** International Autonomous Competition Top 10th Globally — ***July 2026***
- **Visited Qatar Shell eco marathon for racing cars —** *1st Place in Mena Region and 4th Globally* — ***Feb 2026***
- **TCCD Research Day Challenge —** *First Place* — ***May 2025***
- **Line Follower Car Competition, Cairo University Eco Racing team**— *First Place* — ***Feb 2025***
- **NASA Space Apps Challenge, Port Said Branch** — *First Place & Global Nominee* — ***Oct 2024***

Courses

- **IEEE:** Power Electronics Design essentials — ***June 2026***
- **NTI:** Digital Design and Verification using Verilog, VHDL — ***Aug 2025***
- **Sprints Microsoft:** AI and Machine Learning — ***July 2025***
- **ITI-Mahra-Tech:** Embedded C Hardware Essentials and Device Driver Development — ***July 2025***
- **Sprints Microsoft:** Cybersecurity Summer Camp — ***July 2025***
- **IEEE:** introduction to Digital Design Sessions Using Verilog on Questasim — ***June 2025***
- **CS50:** Web Programming with Python and JavaScript — ***Dec 2022***

Technical Projects

- **Autonomous Racing Car Control System** — ***June 2026***
  Developed two custom STM32/ESP32 PCBs for telemetry and vehicle control.
  Transmitted GPS, IMU, RPM, and vehicle data to an NVIDIA Jetson for AI-based autonomous driving.
  Implemented bidirectional communication between the Jetson and embedded controllers to execute steering, throttle, and braking commands.
- **Designing and implementing an embedded smart medical watch** — ***Mar 2026***
  Developed an STM32/ESP32-based wearable embedded system integrating I2C sensors (MAX30102, BMI160, temperature).
  Implemented BLE communication for real-time data streaming to mobile application, while continuous integration using Git.
  Optimized low-power wake-up strategy with a charging Circuit using USB to improve battery efficiency.
- **Designing Motor Control PCB using Altium** — ***Mar 2026***
  Designing a 100cm^2 PCB including ESP32 connected to several ICs like level shifters and Motor Driv8825 to control motors at a known high current and voltage for the 3phase motors to be implemented in a real racing car for motor controlling.
- **Designing and implementation of a telemetry system in a racing car** — ***Jan 2026***
  Implementation of an embedded System using GPS-NEO6M and its antenna and some another sensors like Gyro and acceleration all in one ESP32 and STM32 using their IDEs like HAL Software in CubeIDE. And sending Data using MQTT to an external Host in a web dashboard for Racing Cars for Engineers and drivers. It is tested in Qatar Shel Eco Marathon 2026.

- **Designing Injection Locked and E-TSPC Digital and Analog Frequency Dividers using Cadence Virtuoso** ***Nov 2025***
  Designing and tuning Analog Injection locked in Cadence Virtuoso for achieving the best bandwidth and power consumption for that IC using NMOS and PMOS.
- **Designing LDO and OTA using Cadence Virtuoso** ***Dec 2025***
  Designing and tuning LDO Circuit with voltage-controlled steps potentiometer in Cadence Virtuoso for achieving the best constant voltage and power consumption for that IC using NMOS and PMOS.

- **Developed and deployed a personal portfolio website to showcase my projects and skills** ***Aug 2025***
  - Built with **React (or Next.js)**, responsive design (HTML, CSS, JS), and deployed on **Railway / Vercel / GitHub Pages**
- **Desinging Communication Protocol (I^2C , UART) – NTI Verilog Summer Training Project** ***Aug 2025***
  - Designed and simulated 2 examples of Communication Protocols like I2C , UART and learning the Concepts of these protocols Modules Design and properties
- **8-Bit Processor Designer – NTI VHDL Summer Training Project** ***Aug 2025***
  - Designed and simulated a 8-bit single-cycle simple processor in VHDL, implementing a full datapath, control unit, memory modules, and instruction execution flow using ModelSim and Vivado.
- **RISC-V Processor Designer – IEEE Digital Design Sessions** ***July 2025***
  - Designed and simulated a 32-bit single-cycle RISC-V processor in Verilog, implementing a full datapath, control unit, memory modules, and instruction execution flow using ModelSim and Vivado.
- **Sunlight Simulation and Illumination Analysis on 3D Human Model using MATLAB** ***May 2025***
  - Simulated sun movement over a 3D STL model of a colleague in MATLAB, analyzing surface lighting and shadow effects. Learned 3D design integration, and dynamic visualization, and importing my MATH Models in that simulation.
- **Logic Gates using Transistors and Diodes** ***Apr 2025***
  - Designing Transistors and MOSFETS and Diodes circuits to make simple Logic Gates and implementing them into a real circuit like: 2Bit-Binary Adder/Subtractor.
- **Automated Lane Follower Car** ***Feb 2025***
  - Designed and developed an autonomous car by the STM32 "Blue Pill" microcontroller, programmed using STM32CubeIDE.
- **Designing Multimeter** ***Dec 2024***
  - Developed and Designed a Circuit using Arduino and Op Amps for amplifying and reading All the main circuit Components (Voltage and Current and Ohmic Resistance) in a small Designed PCB
- **Exoplanet Simulation Website** ***Oct 2024***
  - Collaborated to design a website simulating exoplanets using HTML, CSS, JS, and React, completing it in 48 hours for the NASA Space Apps Challenge.

SKILLS

- **Personal Skills:** Leadership, Communication, Time Management.
- **Languages:** Fluent in English and Arabic.
- **communication skills:** Leading and management skills as an class representative.
- **Core Embedded software Skills:**
  - Embedded C, STM32 HAL, FreeRTOS
  - Communication Protocols: UART, I2C, SPI, BLE
  - Debugging: Oscilloscope, Logic Analyzer
  - Version Control Systems: Git (Professionally worked several repositories)
  - Testing & Validation (Unit Testing – Basic)
  - MISRA-C (Learning / Familiar)
  - Scripting (Learning / Familiar)
  - Technical Documentation and software documentation: Professional in word and Canva.
  - Machine Learning: Python, linear regression model